fn cdp_epsilon

Michael Shoemate

July 9, 2024

Proves soundness of `fn cdp_epsilon` in `cdp_epsilon.rs` at commit 0b8f4222 (outdated[1]). This proof is an adaptation of subsection 2.3 of [CKS20].

1 Bound Derivation

Definition 1.1. (Privacy Loss Random Variable). Let $M : \mathcal{X}^n \to \mathbb{Y}$ be a randomized algorithm. Let $x, x' \in \mathcal{X}^n$ be neighboring inputs. Define $f : \mathcal{Y} \to \mathbb{R}$ by $f(y) = log\left(\frac{\mathbb{P}[M(x)=y]}{\mathbb{P}[M(x')=y]}\right)$. Let $Z = f(M(x))$, the privacy loss random variable, denoted $Z \leftarrow PrivLoss(M(x)||M(x'))$.

Lemma 1.2. [CKS20] Let $\epsilon, \delta \geq 0$. Let M: $\mathcal{X}^n \to \mathcal{Y}$ be a randomized algorithm. Then M satisfies (ϵ, δ)-differential privacy if and only if

$$\delta \geq \mathop{\mathbb{E}}_{Z \leftarrow PrivLoss(M(x)||M(x'))}[max(0, 1 - e^{\epsilon - Z})] \tag{1}$$

$$\tag{2}$$

for all $x, x' \in \mathcal{X}^n$ differing on a single element.

Proof. Fix neighboring inputs $x, x' \in \mathcal{X}^n$. Let $f : \mathcal{Y} \to \mathbb{R}$ be as in 1.1. For notational simplicity, let $Y = M(x)$, $Y' = M(x')$, $Z = f(Y)$ and $Z' = -f(Y')$. This is equivalent to $Z \leftarrow PrivLoss(M(x)||M(x'))$. Our first goal is to prove that

$$\sup_{E \subset \mathcal{Y}} \mathbb{P}[Y \in E] - e^{\epsilon}\mathbb{P}[Y' \in E] = \mathbb{E}[max\{0, 1 - e^{\epsilon - Z}\}]. \tag{3}$$

For any $E \subset \mathcal{Y}$, we have

$$\mathbb{P}[Y' \in E] = \mathbb{E}[\mathbb{I}[Y' \in E]] = \mathbb{E}[\mathbb{I}[Y \in E]e^{-f(Y)}]. \tag{4}$$

This is because $e^{-f(y)} = \frac{\mathbb{P}[Y=y]}{\mathbb{P}[Y'=y]}$.

Thus, for all $E \subset \mathcal{Y}$, we have

$$\mathbb{P}[Y \in E] - e^{\epsilon}\mathbb{P}[Y' \in E] = \mathbb{E}\left[\mathbb{I}[Y \in E](1 - e^{\epsilon - f(Y)})\right] \tag{5}$$

Now it is easy to identify the worst event as $E = \{y \in \mathcal{Y} : 1 - e^{\epsilon - f(y)} > 0\}$. Thus

$$\sup_{E \subset Y} \mathbb{P}[Y \in E] - e^{\epsilon}\mathbb{P}[Y' \in E] = \mathbb{E}\left[\mathbb{I}[1 - e^{\epsilon - f(Y)} > 0](1 - e^{\epsilon - f(Y)})\right] = \mathbb{E}[max\{0, 1 - e^{\epsilon - Z}\}] \tag{6}$$

□

[1] See new changes with `git diff 0b8f4222..c83fac2 rust/src/combinators/measure_cast/zCDP_to_approxDP/cdp_epsilon.rs`

Theorem 1.3. [CKS20] Let $M : \mathcal{X}^n \to \mathcal{Y}$ be a randomized algorithm. Let $\alpha \in (1, \infty)$ and $\epsilon \geq 0$. Suppose $D_\alpha(M(x)||M(x')) \leq \tau$ for all $x, x' \in \mathcal{X}^n$ differing in a single entry.[2] Then M is (ϵ, δ)-differentially private for

$$\delta = \frac{e^{(\alpha-1)(\tau-\epsilon)}}{\alpha - 1}\left(1 - \frac{1}{\alpha}\right)^\alpha \tag{7}$$

Proof. Fix neighboring $x, x' \in \mathcal{X}^n$ and let $Z \leftarrow PrivLoss(M(x)||M(x'))$. We have

$$\mathbb{E}[e^{(\alpha-1)Z}] = e^{(\alpha-1)D_\alpha(M(x)||M(x'))} \leq e^{(\alpha-1)\tau} \tag{8}$$

By 1.2, our goal is to prove that $\delta \geq \mathbb{E}[max\{0, 1 - e^{\epsilon - Z}\}]$. Our approach is to pick $c > 0$ such that $max\{0, 1 - e^{\epsilon-Z}\} \leq ce^{(\alpha-1)z}$ for all $z \in \mathbb{R}$. Then

$$\mathbb{E}[max\{0, 1 - e^{\epsilon-Z}\}] \leq \mathbb{E}[ce^{(\alpha-1)z}] \leq ce^{(\alpha-1)\tau}. \tag{9}$$

We identify the smallest possible value of c:

$$c = \sup_{z\in\mathbb{R}} \frac{max\{0, 1 - e^{\epsilon-z}\}}{e^{(\alpha-1)z}} = \sup_{z\in\mathbb{R}} e^{z-\alpha z} - e^{\epsilon-\alpha z} = \sup_{z\in\mathbb{R}} f(z) \tag{10}$$

where $f(z) = e^{z-\alpha z} - e^{\epsilon-\alpha z}$. We have

$$f'(z) = e^{z-\alpha z}(1-\alpha) - e^{\epsilon-\alpha z}(-\alpha) = e^{-\alpha z}(\alpha e^\epsilon - (\alpha-1)e^z) \tag{11}$$

Clearly $f'(z) = 0 \iff e^z = \frac{\alpha}{\alpha-1}e^\epsilon \iff z = \epsilon - log(1 - 1/\alpha)$. Thus

$$\begin{aligned} c &= f(\epsilon - log(1 - 1/\alpha)) && (12)\\ &= \left(\frac{\alpha}{\alpha-1}e^\epsilon\right)^{1-\alpha} - e^\epsilon\left(\frac{\alpha}{\alpha-1}e^\epsilon\right)^{-\alpha} && (13)\\ &= \left(\frac{\alpha}{\alpha-1}e^\epsilon - e^\epsilon\right)\left(\frac{\alpha}{\alpha-1}e^{-\epsilon}\right)^\alpha && (14)\\ &= \frac{e^\epsilon}{\alpha-1}\left(1-\frac{1}{\alpha}\right)^\alpha e^{-\alpha\epsilon}. && (15)\end{aligned}$$

Thus

$$\mathbb{E}[max\{0, 1 - e^{\epsilon-Z}\}] \leq \frac{e^\epsilon}{\alpha-1}\left(1-\frac{1}{\alpha}\right)^\alpha e^{-\alpha\epsilon}e^{(\alpha-1)\tau} = \frac{e^{(\alpha-1)(\tau-\epsilon)}}{\alpha-1}\left(1-\frac{1}{\alpha}\right)^\alpha = \delta \tag{16}$$

□

Corollary 1. [CKS20] Let $M : \mathcal{X}^n \to \mathcal{Y}$ be a randomized algorithm. Let $\alpha \in (1, \infty)$ and $\epsilon \geq 0$. Suppose $D_\alpha(M(x)||M(x')) \leq \tau$ for all $x, x' \in \mathcal{X}^n$ differing in a single entry. Then M is (ϵ, δ)-differentially private for

$$\epsilon = \tau + \frac{\ln(1/\delta) + (\alpha-1)\ln(1-1/\alpha) - \ln(\alpha)}{\alpha-1} \tag{17}$$

Proof. This follows by rearranging 1.3. □

Corollary 2. Let $M : \mathcal{X}^n \to \mathcal{Y}$ be a randomized algorithm satisfying ρ-concentrated differential privacy. Then M is (ϵ, δ)-differentially private for any $0 < \delta \leq 1$ and

$$\epsilon = \inf_{\alpha\in(1,\infty)} \alpha\rho + \frac{\ln(1/\delta) + (\alpha-1)\ln(1-1/\alpha) - \ln(\alpha)}{\alpha-1} \tag{18}$$

Proof. This follows from 1 by taking the infimum over all divergence parameters α. □

[2] This is the definition of (α, τ)-Rényi differential privacy.

1.1 Efficient computation of ϵ

From 2, we have

$$\epsilon(\alpha) = \alpha\rho + \frac{\ln(1/\delta) + (\alpha-1)\ln((\alpha-1)/\alpha) - \ln(\alpha)}{\alpha - 1} \tag{19}$$

$$\epsilon'(\alpha) = \rho + \frac{\ln(\alpha\delta)}{(\alpha-1)^2} \tag{20}$$

$$\epsilon''(\alpha) = \frac{2\alpha\ln(\alpha\delta) - \alpha + 1}{(\alpha-1)^3\alpha} \tag{21}$$

Notice the curve is convex so long as

$$\delta < e^{1/2-1/(2\alpha)}/\alpha \tag{22}$$

Otherwise the curve is concave, with a non-negative derivative for any choice of α:

$$\epsilon'(\alpha) = \rho + \frac{\ln(\alpha\delta)}{(\alpha-1)^2} \geq \rho + \frac{\alpha-1}{2\alpha(\alpha+1)^2} \geq 0 \tag{23}$$

We can find the minimizer α_* by conducting a binary search over the interval $(1, \alpha_{max})$, where α_{max} is discovered via exponential search for a positive derivative.

2 Pseudocode

Precondition

- Type `Q` must have trait `Float`.

Implementation

```
def cdp_epsilon(rho: Q, delta: Q) -> Q:
    if rho.is_sign_negative():
        raise "rho must be non-negative"

    if not delta.is_sign_positive():
        raise "delta must be positive"

    if rho.is_zero():
        return 0

    # checks if derivative is positive
    def deriv_pos(a):
        return rho > -log(a * delta) / (a - 1)**2

    # find bounds
    a_min = 1.01
    a_max = 2
    while not deriv_pos(a_max):
        a_max *= 2

    # optimize alpha
    while True:
        diff = a_max - a_min

        a_mid = a_min + diff / _2

        if a_mid == a_max or a_mid == a_min:
            break

```

```
        if deriv_pos(a_mid):
            a_max = a_mid
        else:
            a_min = a_mid

    # back out epsilon
    a_m1 = a_max.inf_sub(_1)

    numer = (a_m1.inf_div(a_max).inf_ln().inf_mul(a_m1)) \
        .inf_sub(a_max.inf_ln()) \
        .inf_add(delta.recip().inf_ln())

    denom = a_max.neg_inf_sub(_1)

    epsilon = a_max.inf_mul(rho).inf_add(numer.inf_div(denom))

    return max(epsilon, 0)
```

Postcondition

Either a valid epsilon is returned or an error is returned.

3 Proof

Theorem 3.1. For any possible setting of ρ and δ, `cdp_epsilon` either returns an error, or an ϵ such that any ρ-differentially private measurement is also (ϵ, δ)-differentially private.

Proof. The code always finds an $\alpha_* \approx \texttt{a_max} \geq 1.01$. Since $\texttt{a_max} \in (1, \infty)$, then by 2, any ρ-differentially private measurement is also $(\epsilon(\texttt{a_max}), \delta)$-differentially private. Define $\epsilon_{cons}(\alpha)$ as a "conservative" function for computing $\epsilon(\alpha)$, where floating-point arithmetic is computed with conservative rounding such that $\epsilon_{cons}(\alpha) \geq \epsilon(\alpha)$ for $\forall \alpha \in (1, \infty)$. Since $\texttt{epsilon} = \epsilon_{cons}(\texttt{a_max}) \geq \epsilon(\texttt{a_max})$, then any $(\epsilon(\texttt{a_max}), \delta)$-differentially private measurement is also $(\texttt{epsilon}, \delta)$-differentially private. □

References

[CKS20] Clément L. Canonne, Gautam Kamath, and Thomas Steinke. The discrete gaussian for differential privacy. *CoRR*, abs/2004.00010, 2020.